

SCQ (Single Correct Type) :

1. If the circles $x^2 + y^2 + (3 + \sin \beta)x + 2 \cos \alpha \cdot y = 0$ and $x^2 + y^2 + 2 \cos \alpha \cdot x + 2cy = 0$ touch each other, then the maximum value of 'c' is

(A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

2. Equation of chord AB of circle $x^2 + y^2 = 2$ passing through the point P(2, 2) such that $\frac{PB}{PA} = 3$ is

(A) $x = 3y$ (B) $x = y$ (C) $y - 2 = \sqrt{3}(x - 2)$ (D) None of these

3. If two different tangents of $y^2 = 4x$ are normal chords to $x^2 = 4by$ then

(A) $|b| > \frac{1}{2\sqrt{2}}$ (B) $|b| < \frac{1}{2\sqrt{2}}$ (C) $|b| > \frac{1}{\sqrt{2}}$ (D) $|b| < \frac{1}{\sqrt{2}}$

4. If locus of mid-point of any normal chord of the parabola :

$$y^2 = 4x \text{ is } x - a = \frac{b}{y^2} + \frac{y^2}{c}$$

where $a, b, c \in \mathbb{N}$, then $(a + b + c)$ equals to :

(A) 5 (B) 8 (C) 10 (D) None of these

MCQ (One or more than one correct) :

5. If $(x - 2)^2 + (y - 2)^2 = 1$, then which of the following is true ?

(A) maximum value of $x + y$ is $4 + \sqrt{2}$ (B) maximum value of $x - y$ is $\sqrt{2}$
 (C) maximum value of xy is $\frac{9 + 4\sqrt{2}}{2}$ (D) minimum value of $x + y$ is $4 - \sqrt{2}$

6. If from (α, β) , two tangents are drawn to circle $x^2 + y^2 = 4$ so that slopes of tangents are in the ratio 1 : 2

and $f(x) = \alpha^2 x^2 + 12x - \frac{\beta^2}{4}$, then

(A) $f(x) > 0 \forall x \in \mathbb{R}$ (B) Locus of (α^2, β^2) is a hyperbola
 (C) least positive integral value of α is 1 (D) eccentricity of locus of (α^2, β^2) is $\sqrt{2}$

7. A circle C_1 of Radius 'b' touches the circle $C : x^2 + y^2 = a^2$ externally and has its centre on positive x-axis. Another circle C_2 of radius 'c' touches the circle C_1 externally and has its centre on positive x-axis. If $a < b < c$ and the three circles have a common tangent, then triplets (a, b, c) can be equal to

(A) (2, 4, 8) (B) (2, 4, 6) (C) (3, 9, 27) (D) (7, 8, 9)

8. Circles are drawn with OA & OB as diameters, where A & B are points of parabola $y^2 = 4x$. These circles meet at P (other than O). m_1 and m_2 are slope of tangents at A & B respectively and m is slope of chord AB, then (given $m_1 + m_2 \neq 0$, A, B are points other than origin and 'O' is origin)
- (A) A, P, B are collinear points (B) m is harmonic mean of m_1 and m_2
 (C) m is arithmetic mean of m_1 and m_2 (D) OP is perpendicular to AB
9. The line $2x - y = 1$ intersect the parabola $y^2 = 4x$ at the points A and B and the normals at A and B intersect each other at the point G. If a third normal to the parabola through G meets the parabola at C then which of the following statement(s) is/are correct.
- (A) sum of the abscissa and ordinate of the point C is -1 .
 (B) the normal at C passes through the lower end of the latus rectum of the parabola.
 (C) centroid of the triangle ABC lies at the focus of the parabola.
 (D) normal at C has the gradient -1 .

Paragraph Type Question :

Paragraph for Question Nos. 10 to 11

Let $C : x^2 + y^2 - 4x - 6y - 3 = 0$ is a circle and S is a family of circles passing through two fixed points A(3, 7) and B(6, 5).

10. The chords in which the circle C cuts the member of the family S are concurrent at point
- (A) (2, 3) (B) $\left(2, \frac{23}{3}\right)$ (C) $\left(3, \frac{23}{2}\right)$ (D) (3, 2)
11. Equation of member of the family S that bisects the circumference of circle C is
- (A) $x^2 + y^2 - 5x - 1 = 0$ (B) $x^2 + y^2 - 5x + 6y - 1 = 0$
 (C) $x^2 + y^2 - 5x - 6y - 1 = 0$ (D) $x^2 + y^2 + 5x - 6y - 1 = 0$

Paragraph for Question Nos. 12 to 14

Consider the following lines :

$$L_1 : x - y - 1 = 0$$

$$L_2 : x + y - 5 = 0$$

$$L_3 : y - 4 = 0$$

Let L_1 is axis to a parabola, L_2 is tangent at the vertex to this parabola and L_3 is another tangent to this parabola at some point P.

Let 'C' be the circle circumscribing the triangle formed by tangent and normal at point P and axis of parabola. The tangent and normals at the extremities of latus rectum of this parabola forms a quadrilateral ABCD.

12. The equation of the circle 'C' is :
- (A) $x^2 + y^2 - 2x - 31 = 0$ (B) $x^2 + y^2 - 2y - 31 = 0$
 (C) $x^2 + y^2 - 2x - 2y - 31 = 0$ (D) $x^2 + y^2 + 2x + 2y = 31$
13. The given parabola is equal to which of the following parabola ?
- (A) $y^2 = 16\sqrt{2}x$ (B) $x^2 = -4\sqrt{2}x$ (C) $y^2 = -\sqrt{2}x$ (D) $y^2 = 8\sqrt{2}x$

14. The area of the quadrilateral ABCD is :
 (A) 16 (B) 8 (C) 64 (D) 32

Numerical Based Questions :

15. A circle is inscribed (i.e. touches all four sides) into a rhombus ABCD with one angle 60° . The distance from the centre of the circle to the nearest vertex is equal to 1. If P is any point of the circle, then $|PA|^2 + |PB|^2 + |PC|^2 + |PD|^2$ is equal to:
16. Normal at the point P to the parabola $y^2 = 4x$, intersects the circle with SP as diameter at Q also. If $PQ = 2$ units (given that point S is focus of the given parabola), then find the abscissa of point P.



ANSWERKEY OF CAPS – 9+

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|----------|---------|----------|----------|-----------|
| 1. (B) | 2. (B) | 3. (B) | 4. (B) | 5. (ABCD) |
| 6. (BCD) | 7. (AC) | 8. (ABD) | 9. (ABC) | 10. (B) |
| 11. (C) | 12. (A) | 13. (D) | 14. (C) | 15. 11 |
| 16. 3 | | | | |